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METHODS AND APPARATUSES FOR AUTOMATIC GAIN CONTROL OF AN AUDIO SIGNAL

Abstract

Methods, apparatuses, and algorithms for the automatic control of gain applied to, or associated with, an audio signal are described herein. An example method may comprise generating a first amplitude measurement of an audio signal, generating a second amplitude measurement of the audio signal, determining a first difference between the first amplitude measurement and the second amplitude measurement; generating, based on the first difference, a first indication; calculating, based on at least the first indication, a target gain; and modifying, based on the target gain, a gain state associated with the first audio signal according to a first ramp speed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/559,262, filed Feb. 29, 2024, which is incorporated by reference in its entirety.

FIELD

[0002] Aspects described herein generally relate to audio signal processing, and/or hardware and/or software related thereto. More specifically, one or more aspects described herein provide for the automatic control of gain applied to a microphone signal.

BACKGROUND

[0003] In many instances, podcasters, broadcasters, audio streamers, recording artists, sound engineers, and the like may wish to have a stable microphone signal output level irrespective of whether the user is talking softly or loudly into the microphone to minimize unnatural voice fluctuations and pumping of background noise. Background noise pumping may refer to unpredictable (and often drastic) changes in the gain applied to an audio signal as a result of the background noise triggering signal processors to adjust the gain. Users may also wish to utilize automatic gain control solutions that are not microphone (or hardware) dependent.

SUMMARY

[0004] The following presents a simplified summary of the disclosure in order to provide a basic understanding of some aspects of the disclosure. This summary is not an extensive overview of the disclosure. It is not intended to identify key or critical elements of the invention or to delineate the scope of the invention. The following summary merely presents some concepts of the disclosure in a simplified form as a prelude to the more detailed description provided below.

[0005] As described in more detail herein, this application sets forth apparatuses, methods, and algorithms for automatically calculating a gain value to apply to an audio system, which may include one or more microphones, to meet a desired output level. This application further sets forth apparatuses, methods, and algorithms for automatically estimating microphone sensitivity and offsetting said sensitivity.

[0006] An example apparatus may comprise an audio detector. The audio detector may be configured to receive a first audio signal having a first signal level. The audio detector may comprise one or more envelope followers and comparators. The one or more envelope followers may comprise a plurality of envelope followers that may comprise a first envelope follower, a second envelope follower, and a third envelope follower. The first envelope follower may be configured to generate, from a first audio signal and based on a first time constant, a first amplitude measurement. The second envelope follower may be configured to generate, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement. The third envelope follower may be configured to generate, from the first audio signal and based on a third time constant different from the first time constant and the second time constant, a third amplitude measurement. A first comparator may be configured to output a difference between the outputs of the first and second envelope followers. A second comparator may be configured to generate a first indication of whether the output of the first comparator satisfies a first threshold value. A third comparator may be configured to output a difference between the outputs of the first and third envelope followers. A fourth comparator may be configured to generate a second indication of whether the output of the third comparator satisfies a threshold value. The apparatus may comprise a gain control subsystem in communication with the audio detector. The gain control subsystem may comprise a target gain calculation module and gain state update module. The target gain calculation module may be configured to calculate, based on at least one or more of the first and second indications, a target gain for the audio signal. The gain state update module may be configured to modify, based on the target gain, a gain state applied to, or associated with, the audio signal.

[0007] An example method may comprise generating, by a computing device, a first amplitude measurement of an audio signal based on a first time constant; generating, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement; generating, from the first audio signal and based on a third time constant different from the first and second time constants, a third amplitude measurement; outputting a first difference between the first amplitude measurement and the second amplitude measurement; generating a first indication of whether the first difference satisfies a first threshold value; outputting a second difference between the first amplitude measurement and the third amplitude measurement; generating a second indication of whether the second difference satisfies a second threshold value; calculating, based on at least one or more of the first and second indications, a target gain for the first audio signal; and modifying, based on the target gain, a gain state of the system to apply to the audio signal. The method may further comprise modifying the gain state based on a hysteresis of the first target gain. The method may further comprise modifying the gain state of the first audio signal according to a ramp speed. The ramp speed may be based on a difference between the target gain and gain state.

[0008] These as well as other novel advantages, details, examples, features and objects of the present disclosure will be apparent to those skilled in the art from following the detailed description, the attached claims and accompanying drawings, listed herein, which are useful in explaining the concepts discussed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals reference similar elements.

[0010] FIG. **1** illustrates an example network architecture that may be used to implement one or more illustrative aspects described herein.

[0011] FIG. **2** illustrates an example system architecture that may be used to implement one or more illustrative aspects described herein.

[0012] FIG. **3***a* illustrates a block diagram of an example automatic gain control module.

[0013] FIG. **3***b* illustrates a block diagram of a subsystem of the example automatic gain control module of FIG. **3***a*.

[0014] FIG. **3***c* illustrates an example input/output curve of the example automatic gain control module of FIG. **3***a*.

[0015] FIG. **4** illustrates an example hysteresis gain ramp speed plot.

[0016] FIG. **5** illustrates an example flow chart of a method that may be performed to implement one or more illustrative aspects described herein.

[0017] FIG. **6** illustrates another example gain control module.

DETAILED DESCRIPTION

[0018] In the following description of the various examples, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration various examples in which aspects may be practiced. References to “embodiment,” “example,” and the like indicate that the embodiment(s) or example(s) of the invention so described may include particular features, structures, or characteristics, but not every embodiment or example necessarily includes the particular features, structures, or characteristics. Further, it is contemplated that certain embodiments or examples may have some, all, or none of the features described for other examples. And it is to be understood that other embodiments and examples may be utilized and structural and functional modifications may be made without departing from the scope of the present disclosure.

[0019] Unless otherwise specified, the use of the serial adjectives, such as, “first,” “second,” “third,” and the like that are used to describe components, are used only to indicate different components, which can be similar components. But the use of such serial adjectives is not intended to imply that the components must be provided in given order, either temporally, spatially, in ranking, or in any other way.

[0020] Also, while the terms “front,” “back,” “side,” and the like may be used in this specification to describe various example features and elements, these terms are used herein as a matter of convenience, for example, based on the example orientations shown in the figures and/or the orientations in typical use. Nothing in this specification should be construed as requiring a specific three dimensional or spatial orientation of structures in order to fall within the scope of the claims.

[0021] FIG. **1** illustrates an example of a network architecture that may be used to implement one or more illustrative aspects described herein in a standalone and/or networked environment. Device **100** may be a microphone, including any number of microphone types, such as a condenser microphone (e.g., including large-and small-diaphragm and electret condenser), a dynamic microphone (e.g., including moving coil and ribbon microphones), or a MEMS microphone, among others. Device **100** may be a smartphone microphone (such as that of smartphone **104**), a desktop or laptop microphone (such as that of device **102**), a headset microphone, a two-way radio microphone, or any other microphone that may be connected to and/or in communication with a device (such as devices **102** and **104**). Device **100** may be a component of a wireless microphone system, such as, for example, a wireless transmitter, a wireless receiver, or a wireless transceiver. Device **100** may be any number of audio interfaces. Any one or more of devices **100**, **102**, **104**, and **106** may be any type of known computer or server. Device **102** may be a desktop computer. Device **104** may be a smartphone or tablet. Devices **102** and **104** may include a user interface, including a graphical user interface, to allow a user to interact with the system. Device **106** may be a data server, including a cloud-based data server. Devices **100**, **102**, **104**, and **106** may be interconnected via wide area network (WAN), such as the Internet. Other networks may also or alternatively be used, including local area networks (LAN), wireless networks, personal networks (PAN), and the like. Devices **100**, **102**, **104**, and **106** and other devices (not shown) may or might not be communicatively connected to one or more of the networks via twisted pair wires, coaxial cable, fiber optics, radio waves, or other communication media. Device **100** may be communicatively connected to device **102** and/or device **104** via connections **108***a* and/or **108***b*, respectively. Device **102** may be communicatively connected to device **106** via connection **110** and device **104** may be communicatively connected to device **106** via connection **112**.

[0022] FIG. **2** illustrates an example of a system architecture that may be used to implement one or more illustrative aspects described herein. One or more of automatic gain control modules **200**, **200***a*, **200***b*, **200***c*, **200***n* (hereinafter collectively referred to as “the gain control modules”) (gain control modules **200***a*, **200***b*, **200***c*, **200***n* described in greater detail in FIG. **6**) may be implemented as (or controllable by) a processor **204**. Individual reference to gain control module **200** and/or **200***a*, **200***b*, **200***c*, **200***n* may also apply to any gain control module(s) not explicitly referenced. For the sake of simplicity, and except where otherwise noted, reference to gain control module **200** shall include gain control modules **200***a*, **200***b*, **200***c*, **200***n*, and vice versa.

[0023] Processor **204** may be communicatively connected to memory **203**. The memory **203** may store operating system software **206** for controlling overall operation of the gain control modules and/or control logic **207** for instructing the gain control modules to perform aspects described herein. Some or all of the aspects described herein may be performed by firmware applications intended to run without an operating system, such as, for example, an ASIC, or a DSP chip configured to perform the operations described natively or by the control of an operating system, etc. Functionality of the control logic **207** may refer to operations or decisions made automatically based on rules coded into the control logic **207**, made manually by a user providing input into the system, and/or a combination of automatic processing based on user input (e.g., queries, data updates, user-selected modes, a list of input devices previously setup with the software application, etc.). Memory **203** may store data used in performance of one or more aspects described herein, including at least one database **208**. Memory may also store other data. For example, where the memory **203** is part of, for example, the input device **212**, such as device **100**, the memory may store its operating system and/or the software application that performs aspects described herein, user preferences such as preferred modes, a list of input devices (such as device **100**, among others) previously setup with the software application, communication protocol settings, and/or data supporting any other functionality of the input device **212**.

[0024] One or more aspects may be embodied in computer-usable or readable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices as described herein, such as processor **204**. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other device. The modules may be written in a source code programming language that is subsequently compiled for execution, such as, for example, C, C++, and/or Assembly, or may be written in a scripting language such as (but not limited to) Python, Perl, PHP, Ruby, JavaScript, and the like. The computer executable instructions may be stored on a computer readable medium such as a nonvolatile storage device. Any suitable computer readable storage media may be utilized, including hard disks, CD-ROMs, optical storage devices, magnetic storage devices, solid state storage devices, and/or any combination thereof. In addition, various transmission (non-storage) media representing data or events as described herein may be transferred between a source and a destination in the form of electromagnetic waves traveling through signal-conducting media such as metal wires, optical fibers, and/or wireless transmission media (e.g., air and/or space). Various aspects described herein may be embodied as a method, a data processing system, or a computer program product. Therefore, some or all of the functionalities performed by the gain control modules may be embodied in whole or in part in software, firmware, and/or hardware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects described herein, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein.

[0025] With further reference to FIGS. **1** and **2**, one or more of the gain control modules may be embodied in any one or more of devices **100**, **102**, **104**, and/or **106**, as well as (or alternatively) in one or more additional devices (not shown). One or more of the gain control modules, device controller **201**, memory **203**, processor **204**, output device **214**, and/or input device **212** may be embodied in any one or more of devices **100**, **102**, **104**, and/or **106**, as well as (or alternatively) in one or more additional devices (not shown). Aspects described herein may be operational with numerous other general purpose and/or special purpose computing system environments or configurations. Examples of other computing systems, environments, and/or configurations that may be suitable for use with aspects described herein include, but are not limited to, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network personal computers (PCs), minicomputers, mainframe computers, supercomputers configured to run online application programming interfaces (APIs), distributed computing environments that include any of the above systems or devices, and the like. The gain control modules may be implemented as embedded software running in, for example, device **100**, and executable on processor **204**. The gain control modules may be implemented as an external signal processor, such as a hardware DSP module, a real-time software processor, an offline software processor, or a software plug-in (including VST, AU, and AAX formats). The gain control modules may include several states, including an always-active state and an on/off state based on a user's input. In the always-active state, the gain control modules may be implemented as a VST plugin. The VST may be inserted into a digital audio workstation ("DAW") to actively monitor an incoming audio signal. One or more of the gain control modules may be implemented as software or a plugin compatible with one or more video communications platforms, video streaming platforms, audio communications platforms, and/or audio streaming platforms.

[0026] As shown in FIG. **2**, input device **212** (such as device **100**) may be in communication with device controller **201**. Device controller **201** may facilitate interaction from input device **212** to one or more of the gain control modules. Analog and/or digital audio may be transmitted from input device **212** to device controller **201**. Digital data may be transmitted bidirectionally (from input device **212** to device controller **201**, and/or from device controller **201** to input device **212**). Input device **212** may include, for example, one or more universal serial bus (USB) connectors, one or more XLR connectors, one or more power connectors, and/or any other type of data and/or power connectors suitable for transporting signals such as power, digital data (including digital audio signals), and/or analog audio signals to and from input device **212**. Where the connection is wired, device controller **201** may further comprise a data interface (not shown) for communicating with input device **212**. For example, the data interface may comprise a USB interface or an XLR interface. While several wired connections are discussed between device controller **201** and input device **212**, other types of wired or wireless connections may be used. For example, the connection between device controller **201** and input device **212** may instead be a wireless connection, such as a Wi-Fi connection or other proprietary wireless connection protocols, a Bluetooth connection, a near-field connection (NFC), and/or an infrared connection. Where the connection is wireless, device controller **201** and input device **212** may include a wireless communications interface.

[0027] As has been discussed, users may wish to have a stable microphone signal output level irrespective of whether the user is talking softly or loudly into the microphone to minimize unnatural voice fluctuations and pumping of background noise. Users may also wish to utilize automatic gain control solutions that are not microphone (or hardware) dependent. At least some problems may arise with some gain control algorithms, such as, for example, drastic changes in the applied gain when there has not been a significant change in user activity. These drastic changes in gain may result in unwanted expansion behavior and, as previously mentioned, background noise (e.g., noise floor) pumping. Other shortcomings with some gain control algorithms may arise, such as, for example, deficiencies in fully matching the input level signal to a desired output level, a hardware dependent gain input/output ("I/O") curve (which may lead to varying performance of the algorithm based on the microphone the user is talking into), and an overly active limiter (which may distort the audio signal). As described herein, automatic gain control may be improved with an automatic gain control module that may be configured to, for example: 1) appropriately adjust the gain applied to, or associated with, an audio system by a sufficient amount that prevents expansion-like behavior and noise floor pumping, by, for example, utilizing a hysteresis to adjust a signal level; 2) consistently achieve a desired output level irrespective of the loudness of a user's voice; 3) exhibit hardware-independence; and 4) estimate the effective sensitivity of a microphone and apply an offset gain to the microphone signal.

[0028] As shown in FIG. **3***a*, gain control module **200** may include audio detector **302** (hereinafter referred to as "detector **302**") and gain control subsystem **304** (hereinafter referred to as "subsystem **304**"). Detector **302** may be configured as many types of detectors, such as, for example, a voice activity detector ("VAD"). Detector **302** may be implemented in any number of manners, such as, for example, an energy-based VAD or a cepstral VAD. Detector **302** may be configured to receive audio signal x(n) and detect, for example, the presence of human speech. Detector **302** may be configured to output one or more indications **334**, **336** to subsystem **304**. Detector **302** is further described in FIG. **3***b*.

[0029] Subsystem **304** may include comparator **305**. As with comparators **328**, **328**, **330**, **332**, comparator **305** may be embodied in whole or in part in software, firmware, and/or hardware or hardware equivalents such as integrated circuits, an FPGA chip, and the like. Comparator **305** may be configured to receive one or more of indications **334**, **336** from detector **302** and generate an output to target gain calculation module **306** and gain state update module **308** based on the state of one or more of indications **334**, **336**. Comparator **305** may be communicatively coupled with gain state update module **308**. Comparator **305** may generate an output to gain state update module **308** based on the state of one or more of indications **334**, **336**. Indications **334** and **336** may generally instruct subsystem **304** to recalculate the target gain. Indication **336** may generally instruct subsystem **304** to adjust a gain state to move the gain state toward an incumbent target gain or a recalculated target gain. Indications **334**, **336** may be represented as Boolean data (e.g., having a "true" state and a "false" state, or a value of "1" or "0", respectively).

[0030] Indications **334**, **336** may be represented as other forms of data, such as, for example, continuous quantitative variables, multi-level logic, a voltage, a pulse signal, etc. Indications **334**, **336** may be represented as other forms of data that generally indicate whether one or more conditions, etc., have been met. Comparator **305** may be implemented in various manners, including, for example, as a Boolean operator. Based on the output of comparator **305**, target gain calculation module **306** may recalculate a target gain according to aspects described herein. For example, if both indications **334**, **336** are "true" (e.g., value of "1"), the output of comparator **305** may instruct target gain calculation module **306** to recalculate target gain and gain state update module **308** to adjust the gain state toward the incumbent target gain or recalculated target gain. If indication **334** is "false" (e.g., value of "0"), but indication **336** is true, the output of comparator **305** may instruct gain state update module **308** to adjust the gain state toward the incumbent target gain or recalculated target gain.

[0031] As discussed, subsystem **304** may include target gain calculation module **306** and gain state update module **308** (also referred to as "update module **308**"). Target gain calculation module **306** may include Input/Output Curve **350** (described in greater detail in FIG. **3***c*) (hereinafter referred to as "curve **350**"). Target gain calculation module **306** may be configured to implement curve **350** to calculate the target gain. The structure of curve **350** may be chosen to help reduce the impact of background noise, such as, for example, background noise coming from background talkers.

[0032] Update module **308** may be configured to receive an output from target gain calculation module **306** and an output of indication **336** from detector **302**. Update module **308** may be configured to modify a gain state of the system and/or the rate by which to adjust the gain state of the system toward a target gain. The output of indication **336** from detector **302** may be represented as Boolean data. The output of target gain calculation module **306** may be represented as non-Boolean data, such as, for example, a value represented as decibels (dB), decibels relative to full scale (dBFS), decibel Volt (dBV), sound pressure level (dB SPL), and the like. Update module **308** may be configured to apply one or more smoothing functions to the output of target gain calculation module **306** to help minimize the presence of transient artifacts in the signal that may arise from abrupt changes in gain. The smoothing function of update module **308** may be a linear, exponential, or other non-linear curve between the current gain state of update module **308** and the target gain as determined by target gain calculation module **306**. The output generated by update module **308** may reflect a smooth transition between an "adjust" state and an "inactive" state. Gain state update module **308** may transition between these states any number of times (e.g., for a fixed number of times or indefinitely).

[0033] Automatic gain control module **200** may include compressor **310** and limiter **312**. Compressor **310** may be configured to receive an output signal from update module **308** and perform further processing on the audio signal, such as, for example, compression of the signal x(n). Compressor **310** may be configured to provide a compressed audio signal to limiter **312**. Compressor **310** and/or limiter **312** may be configured with a number of thresholds and ratios.

Limiter **312** may be configured, for example, as a full-band limiter or a multi-band limiter. Limiter **312** may further process the gain-corrected signal x(n) and provide a signal y(n) for further processing and/or for output via an output device such as, for example, output devices **214** and/or **606** (described with respect to FIG. **6**). An audio output device, such as output device **606** (discussed in greater detail with respect to FIG. **6**), may receive output signal y(n) for further processing or for playback.

[0034] FIG. **3***b* illustrates detector **302** in greater detail. Detector **302** may be configured to receive an analog and/or digital signal x(n). Detector **302** may include filter **320**; envelope followers **322**, **324**, **326**; comparators **328**, **329**; and comparators **330**, **332**. Filter **320** may be any number of types of filters, including, but not limited to, minimum phase, linear phase, feed-forward, feed-back, bandpass, etc. Filter **320** may have a passband between 50 and 5000 Hz (or some other frequency range). Filter **320** may have a passband between more or less than 50 Hz (e.g., 1 Hz, 20 Hz, 60 Hz) and more or less than 5000 Hz (e.g., 5500 Hz, 6000 Hz, 7500 Hz, up to the highest-supported frequency of the system, etc.).

[0035] Filter **320** may provide a filtered input signal x(n) to envelope followers **322**, **324**, **326** (collectively referred to as “the envelope followers”). Envelope followers **322**, **324**, **326** may analyze, or measure, the amplitudes, or envelopes, of the filtered signal x(n). The envelope followers may be configured with one or more time resolutions, or time constants. Each time constant may correspond to an attack time, decay, and/or a release time. The envelope followers may be configured with equivalent time constants that correspond to an equivalent attack time, decay, and release time. The envelope followers may be configured with time constants that are different, and, thus, correspond to different attack times, different decays, and different release times. The envelope followers may be configured with time constants that correspond to an attack time of any duration, such as 0.1, 1, 10, 100, or 1,000 milliseconds (ms) (or any other value). The envelope followers may be configured with time constants that correspond to a release time of 0.1, 1, 10, 100, 1,000, or 5,000 ms (or any other value). The envelope followers may be configured with time constants that correspond to a decay rate of any duration, such as 0.1, 1, 10, 100, or 1,000 ms (or any other value). The envelope followers may operate according to any number and/or combination of methods of envelope followers, including single-pole averagers, moving averagers, frame-by-frame averagers, or peak-hold algorithms.

[0036] As has been discussed, the envelope followers may be configured to generate one or more amplitude measurements of signal x(n). The amplitude measurements may represent one or more values, such as, for example, the instantaneous signal input level, a weighted average level, and/or noise floor level of signal x(n). Detector **302** may be configured to compare the difference in amplitudes of the instantaneous signal input and a noise floor level. Detector **302** may generate an indication **334** which may be based on a comparison of the measurements of instantaneous signal input level (which may be generated by envelope follower **322**) and a weighted average level (which may be generated by envelope follower **324**) to, for example, a loudness threshold. The loudness threshold may be a value represented as, for example, dB. The loudness threshold may be used to compare the respective envelopes generated by envelope followers **322**, **324**, which themselves may be generated according to a number of different decay and/or release times. Detector **302** may generate an indication **336** that may be based on a comparison of the measurements of instantaneous signal input level (envelope follower **322**) and the noise floor level (envelope follower **326**) to, for example, an activity threshold. The activity threshold may be a value represented as, for example, decibels (dB), decibels relative to full scale (dBFS), decibel Volt (dBV), or sound pressure level (dB SPL), and may generally signify a user is speaking and that the input signal is not due to inactivity of the microphone.

[0037] The loudness threshold and/or the activity threshold may be updated at various frequencies, such as, for example, dynamically (e.g., in real-time) or according to any number of periodic frequencies. The loudness threshold and/or the activity threshold may be fixed. The loudness threshold and/or activity threshold may be stored in, for example, database **208**.

[0038] The envelope followers may convert the amplitude measurements to a value, for example, represented by decibels (dB), decibels relative to full scale (dBFS), decibel Volt (dBV), or sound pressure level (dB SPL), and may output one or more values to comparators **328** and/or **329**. The output of the envelope followers may be linear or logarithmic.

[0039] In an example, envelope followers **322**, **324** may output amplitude measurements to comparator **328** which may be represented as dB or dBFS. Comparator **328** may subtract the value of the amplitude measurement by envelope follower **324** from that of envelope follower **322** to obtain a value that may represent the relationship between the respective amplitude envelopes generated by envelope followers **324** and **322** (hereinafter generally referred to as an "amplitude relationship"). Comparator **328** may output the amplitude relationship to comparator **330**. In other words, comparator **328** may output a value that represents, for example, the difference between the amplitude measurements of envelope followers **322**, **324**. Envelope followers **322**, **326** may output amplitude measurements to comparator **329** which may be represented as dB or dBFS. Comparator **329** may subtract the value of the amplitude measurement by envelope follower **326** from that of envelope follower **322** to obtain another amplitude relationship. In other words, comparator **329** may output a value that represents, for example, the difference between the amplitude measurements of envelope followers **322**, **326**. Comparator **329** may output the value of the amplitude relationship to comparator **332**. While the above disclosure generally describes the operations of comparators **328**, **329** where the envelope followers are configured as logarithmic envelope followers, it is understood that comparators **328**, **329** may operate with mathematically equivalent behavior where the envelope followers may be configured as linear envelope followers. That is, where reference is made to subtracting one value of an amplitude measurement from that of another (e.g., in the logarithmic domain), it is understood that linear envelope followers would perform division in the linear domain.

[0040] Comparators **330**, **332** may utilize the amplitude relationship values from comparators **328**, **329**, respectively, to generate indications **334**, **336**, respectively. Comparators **330**, **332** may generate indications **334**, **336** by comparing the amplitude relationship values from comparators **328**, **329**, respectively, to one or more thresholds. Comparator **330** may compare the amplitude relationship value output from comparator **328** to, for example, a loudness threshold. If the amplitude relationship value output from comparator **328** is greater than the loudness threshold, comparator **330** may generate indication **334** (e.g., indication **334** is "true" or has a value corresponding to "1"). Comparator **332** may compare the amplitude relationship value output from comparator **329** to, for example, an activity threshold. If the amplitude relationship value is greater than the activity threshold, comparator **332** may generate indication **336**. Based on the state of indications **334**, **336**, comparator **305** may determine whether to issue an instruction for target gain calculation module **306** to recalculate the target gain for the system and/or an instruction for update module **308** to adjust the gain state of the system toward the incumbent target gain value. For example, as discussed above, if indication **336** is "true" (e.g., amplitude relationship value output from comparator **329** is greater than activity threshold) (e.g., value corresponding to "1"), this may indicate that a user is speaking and may instruct gain state update module **308** to adjust the gain state towards the incumbent target gain. If both indication **334** and indication **336** are "true," target gain calculation module **306** may be instructed to recalculate the target gain of the audio system and to adjust the gain state towards the recalculated target gain. If indication **336** is "true," but indication **334** is "false" (e.g., amplitude relationship value output from comparator **328** is less than loudness threshold) (e.g., value corresponding to "0"), gain state update module **308** may be instructed to adjust the gain state of the system toward the incumbent target gain without recalculating the target gain of the system.

[0041] Gain control module **200** may include frame-by-frame averagers (not shown) configured to perform frame-by-frame amplitude detection of an audio signal, such as signal x(n). One or more frame averagers may be configured to capture a root mean square (RMS), peak, and/or other measure of a frame of the audio signal. The frame may be any size, or duration. Frame-by-frame averagers may employ any number of measuring algorithms, such as linear-scaled algorithms, logarithmic-scaled algorithms, and/or algorithms formatted with other scaling. The frame-by-frame averagers may output to comparators **330** and/or **332**. Comparators **330** and/or **332** may utilize one or more of these measurements to determine the values of indications **334**, **336**, respectively, according to aspects described herein.

[0042] FIG. **3***c* illustrates an example input/output (“I/O”) curve **350** (hereinafter referred to as “curve **350**”) that may represent one of many I/O curves of target gain calculation module **306**. Target gain calculation module **306** may be configured to implement curve **350** to calculate the target gain. The structure of curve **350** may be chosen to help reduce the impact of background noise, such as, for example, background noise coming from background talkers. Curve **350** may be configured to help configure a gain I/O curve that might not be hardware dependent (i.e., dependent on the particular microphone) and may help enable consistent performance over a wide range of microphone types. Curve **350** may be modeled as a function of signal input level versus signal output level. The signal output level may reflect an expanded or limited signal level. The signal input level and signal output level may both be represented as values comparable to, for example, decibels relative to full scale (dBFS).

[0043] Line **360** represents a desired output level. Line **362** represents equivalence between the output level and the input level. In other words, line **362** represents instances when 0 dB of gain is applied to the signal. Target gain calculation module **306** may determine target gain by measuring the difference between curve **350** and line **362**. For example, when curve **350** is above line **362**, the calculated target gain to reach desired output level **360** is positive (in decibels). When curve **350** falls below line **362**, the calculated target gain to reach desired output level **360** is negative (in decibels).

[0044] Curve **350** may include multiple states, for example, that may be represented by regions **352**, **354**, **356**. Region **352** may correspond to, for example, an expansion region. If the signal input level falls within region **352**, target gain calculation module **306** may expand the signal output to a predetermined level, as reflected by segment **352***a*. Region **354** may correspond to, for example, a max boost region. If the signal input level falls within region **354**, target gain calculation module **306** may boost the signal output by a maximum predetermined gain, as reflected by segment **354***a*. Region **356** may correspond to, for example, a match desired level region. If the input signal level falls within region **356**, target gain calculation module **306** may calculate a target gain to achieve desired output level. Segments **352***a* and/or **356***a* may be linear or curved (when represented in the logarithmic domain). Segment **354***a* may be linear (when represented in the logarithmic domain). Segment **354***a* may be parallel with line **362**. Segment **356***a* may represent an equivalent value (in dBFS) as that of line **360**. Curve **350** may be segmented into fewer or more divisions, or regions, than regions **352**, **354**, **356**. Segment **352***a* may have a steeper slope than that of segment **354***a*, which may in turn have a steeper slope than that of segment **356***a*.

[0045] FIG. **4** illustrates an example hysteresis gain ramp speed plot **400** (hereinafter referred to as “plot **400**”) that update module **308** may implement to determine the rate, or speed, at which, if necessary, to increase or decrease the gain state of the system until it reaches either an incumbent or a recalculated target gain. Plot **400** may be implemented as or incorporated into, for example, software executable by, for example, processor **204**, firmware, and/or a combination thereof. Update module **308** may utilize plot **400** to adjust the gain of the system (to the extent adjustment is required) in order to reach a calculated target gain. Plot **400** may be represented as a function of ramp speed (line **402**), versus difference between the gain state of the system and target gain (line **403**).

[0046] Elements **402***a* and **402***b*-**402***c* (and corresponding lines **402***b*′, **402***c*′, **402***c*″, **402***d*′, **402***e*′, respectively) of line **402** may denote different ramp speed (or rate) values represented as, for example, decibels per second (dB/Sec), ranging from, for example, zero (**402***a*) to infinity. Lines **402***a*-**402***e* may generally represent, in an increasing manner, various durations of time it takes to increase (or decrease, as the case may be) the gain state of the system to match an incumbent or calculated target gain. Plot **400** may include fewer or more than lines **402***a*-**402***c*. In other words, line **402** of plot **400** may be segmented in any number of divisions from zero to infinity. Element **402***b* and line **402***b*′ may represent, for example, a standard, or default, ramp speed. Element **402***c* and lines **402***c*′, **402***c*″ may represent, for example, a ramp speed with a greater value in, for example, dB/Sec, than element **402***b* and line **402***b*′. Element **402***d* and line **402***d*′ may represent, for example, a ramp speed with a greater value in, for example, dB/Sec, than element **402***c* and lines **402***c*′, **402***c*″, and may correspond to conditions in which a high increase rate is utilized. Element **402***e* and line **402***e*′ may represent, for example a ramp speed with a greater value in, for example, dB/Sec, than element **402***d* and line **402***d*′, and may correspond to conditions in which a high decrease rate is utilized. In other words, the greater the difference between the gain state of the system and the target gain in the negative direction (i.e., from zero towards negative infinity), the greater the value of the increase rate may be. The greater the difference between the gain state of the system and the target gain in the positive direction (i.e., from zero towards infinity), the greater the value of the decrease rate may be.

[0047] Lines **403***a*, **403***a*′, **403***b*, **403***b*′, **404***a*, **404***a*′, **404***b*, **404***b*′ may represent various differences (or distances) between the gain state of the system, such as the gain applied to signal x(n), and a calculated target gain, represented as, for example, decibels (dB), ranging from, for example, negative infinity to zero to infinity. Plot **400** may include fewer or more than lines **403***a*, **403***a*′, **403***b*, **403***b*′, **404***a*, **404***a*′, **404***b*, **404***b*′. In other words, line **403** of plot **400** may be segmented into any number of divisions from negative infinity to infinity.

[0048] Lines **403***a*, **403***a*′, **403***b*, **403***b*′ may represent gain difference thresholds. Lines **404***a*, **404***a*′, **404***b*, **404***b*′ may represent convergence thresholds. Plot **400** may include any number of gain difference thresholds and/or convergence thresholds. In operation, when the difference between the gain state and calculated gain exceeds a gain difference threshold with an increasing magnitude of difference, update module **308** may adjust the ramp rate based on corresponding lines **402***b*′, **402***c*′, **402***c*″, **402***d*′, and **402***e*′. Update module **308** may return to the previous ramp rate when the magnitude of difference between the gain state and calculated gain decreases to less than the corresponding convergence threshold **404***a*, **404***a*′, **404***b*, **404***b*′.

[0049] In operation, update module **308** may measure an initial difference between the gain state of the system and the target gain and apply an appropriate ramp rate. Update module **308** may apply a given ramp rate **402***a*, **402***b*, **402***c*, **402***d*, **402***e* until either 1) the magnitude of the difference decreases below the next convergence threshold that is less than the absolute of the gain difference threshold; or 2) the magnitude of the difference increases to the point of exceeding the next gain difference threshold.

[0050] For example, if the initial difference falls between zero and gain difference threshold **403***a*, update module **308** may apply ramp rate **402***b*′. Update module **308** may continuously measure the difference between the gain state of the system and the target gain. Update module **308** may measure the difference between the gain state of the system and the target gain at any number of frequencies and/or intervals, such as, for example, at various time intervals (e.g., 1 ms, 10 ms, 50 ms, 100 ms, 500 ms, 1000 ms, 10,000 ms, etc.), or continuously. If, for example, update module **308** re-measures the difference, and the new difference exceeds the gain difference threshold **403***a* but does not exceed gain difference threshold **403***b*, update module **308** may apply ramp rate **402***c*′. If, for example, update module **308** re-measures the difference, and the new difference exceeds convergence threshold **404***b* but does not exceed gain difference threshold **403***b*, update module **308** will not apply a new ramp rate (i.e., update module **308** may continue to apply ramp rate **402***c*′).

[0051] If, for example, update module **308** re-measures the difference, and the new difference exceeds gain difference threshold **403***b*, update module **308** may apply ramp rate **402***e*′. After update module **308** re-measures the difference, if the new difference is less than gain difference threshold **403***b* but is still greater than convergence threshold **404***b*, update module will not change the ramp rate. After update module **308** re-measures the difference, if the new difference is less than convergence threshold **404***b* but greater than convergence threshold **404***a*, update module **308** may change the ramp rate to ramp rate **402***c*′. After update module **308** re-measures the difference, if the new difference is less than gain difference threshold **403***a* but is still greater than convergence threshold **404***a*, update module will not apply a new ramp rate (i.e., update module **308** may continue to apply ramp rate **402***c*′). After update module **308** re-measures the difference, if the new difference is less than convergence threshold **404***a* and greater than 0, update module **308** may apply ramp rate **402***b*′. After update module **308** re-measures the difference, if the new difference exceeds convergence threshold **404***a* but does not exceed gain difference threshold **403***a*, update module **308** might not apply a new ramp rate (i.e., update module **308** may continue to apply ramp rate **402***b*′).

[0052] FIG. **5** illustrates an example flow chart of a method **500** that may be performed to implement one or more illustrative aspects described herein. Some or all of the steps may be performed by gain control module **200** housed in, for example, device **100**. Additionally, or alternatively, some or all of the steps may be performed by gain control module **200** housed in a device connected to device **100** (such as devices **102**, **104**, **106**, and/or other devices operating a software application capable of performing the operations described herein). Some or all of the steps may be performed by firmware applications intended to run without an operating system, such as, for example, an application specific integrated circuit (ASIC), or a digital signal processing (DSP) chip configured to perform the operations described herein natively or by the control of an operating system, etc. While the method shows particular steps in a particular order, the method may be further subdivided into additional sub-steps, steps may be combined, the steps may be performed in other orders, and some steps may be omitted without necessarily deviating from the concepts described herein.

[0053] In operation, detector **302** may receive an audio signal x(n) from an input device, such as device **100**. (step **501**, FIG. **5**). The audio signal may be generated by, for example, a user speaking into device **100**. Envelope followers **322**, **324**, **326** may analyze the signal amplitudes (or envelopes) according to aspects described herein (step **503**). Comparators **328**, **329** may respectively determine differences (or relationships) between amplitude measurements, such as, for example, the differences between amplitude measurements generated by envelope followers **322**, **324** and between those generated by envelope followers **322**, **326** (step **505**). Comparators **330**, **332** may compare the amplitude relationship values output by comparators **328**, **329**, respectively, to one or more thresholds as discussed herein (step **507**). Comparators **330**, **332** may generate indications **334**, **336**, respectively, and output them to comparator **305** (step **509**). Comparator **305** may be configured to receive one or more of indications **334**, **336** from comparators **330**, **332**, respectively, and generate an output to target gain calculation module **306** based on the state of one or more of indications **334**, **336**. If indications **334**, **336** are “true” (step **511**: YES), target gain calculation module **306** may recalculate the target gain of the audio system according to aspects described herein (step **513**). In an example, target gain calculation module **306** may recalculate the target gain of the audio system according to aspects described herein if indication **334** is “true.” Processor **204** may instruct update module **308** to determine the difference (or distance) between the gain state of the system and the incumbent or recalculated target gain (step **519**). Update module **308** may determine, based on the difference (or distance) between the gain state of the system and the incumbent (or recalculated) target gain, the appropriate ramp speed by which to increase or decrease the gain state of the system to meet the incumbent (or recalculated) target gain as described herein (step **521**). Update module **308** may utilize methods such as, for example, plot **400**, to determine the appropriate ramp speed. Update module **308** may adjust the signal gain at a given ramp speed as described herein, to meet the incumbent or recalculated target gain (step **523**). Gain control module **200** may receive instructions to automatically continue steps **501** through **523** (step **527**: YES) for an indefinite number of iterations or to automatically continue steps **501** through **523** for a set number of iterations. In an example, gain control module **200** may receive instructions to terminate the procedure **500** (step **527**: NO).

[0054] If one of indications **334**, **336** is “false” (step **511**: NO), but indication **336** is “true” (step **517**: YES), processor **204** may instruct update module **308** to perform step **519** as described above. Update module **308** may perform steps **521** and **523** as described above. Update module **308** may utilize methods such as, for example, plot **400**, to determine the appropriate ramp speed. Gain control module **200** may receive instructions to automatically continue steps **501** through **523** (step **527**: YES) for an indefinite number of iterations or to automatically continue steps **501** through **523** for a set number of iterations. In an example, gain control module **200** may receive instructions to terminate the procedure **500** (step **527**: NO).

[0055] If neither the first nor the second amplitude relationships satisfy respective threshold values (i.e., neither indications **334** nor **336** are “true”), processor **204** may instruct automatic gain control module **200** to maintain the current gain state of the system (step **517**: NO). Gain control module **200** may receive instructions to automatically continue steps **501** through **523** (step **529**: YES) for an indefinite number of iterations or to automatically continue steps **501** through **523** for a set number of iterations. In an example, gain control module **200** may receive instructions to terminate the procedure **500** (step **529**: NO).

[0056] Any of the circuitry in FIGS. **3***a*-**3***b* may be implemented as or incorporated into, for example, as a microcontroller, a programmable gate array (PGA), as a MOS integrated circuit (IC) chip, an ASIC, a DSP chip configured to perform the operations described herein natively or by the control of an operating system, a complex programmable logic device (CPLD), an FPGA chip, or an analog electrical circuit. The ASIC could contain a transistor, such as a FET. Any of the operations described herein may be implemented with hardware, software executable by, for example, processor **204**, firmware, and/or a combination thereof.

[0057] The aspects described herein may be performed by a number of devices and/or device configurations. The aspects describe herein may be performed by device **100**. No other equipment might be necessary to perform the operations described herein. A user may connect device **100** to devices **102**, **104**, **106**, and/or other devices operating a software application capable of performing the operations described herein. Some or all of the components of FIGS. **3***a*-**3***c*, **4**, and **6** may be logical blocks implemented as embedded software running in, for example, device **100** and/or other devices operating a software application capable of performing operations described herein, and may be executable by processor **204**. The aspects described herein may be performed by firmware applications intended to run without an operating system, such as, for example, an ASIC, or a DSP chip configured to perform the operations described herein natively or by the control of an operating system, etc. In an example, the aspects described herein can be performed by a smartphone, desktop computer, laptop computer, and/or other devices that may or might not have an internal microphone and/or a software application capable of performing the operations described herein. No other audio equipment might be necessary to perform the operations described herein.

[0058] FIG. **6** illustrates an example block diagram whereby one or more input devices **100**, **600**, **602**, and **604** may be connected to gain control module modules **200***a*, **200***b*, **200***c*, and **200***n*, respectively, operating in concert with auto-mixer **605**. Gain control modules **200***a*, **200***b*, **200***c*, and **200***n* (collectively referred to as “the gain control modules”) may operate in accordance with concepts described herein. The input devices may include device **100** and microphones **600**, **602**, **604**. Microphones **600**, **602**, **604** may include any number of microphone types, including a condenser microphone (including large-and small-diaphragm and electret condenser), a dynamic microphone (including moving coil and ribbon microphones), a MEMS microphone, etc. The input devices may include more or fewer microphones. The input devices may connect to the gain control modules using any one of a variety of different connectors, including a LEMO connector, an XLR connector, a TQG connector, a TRS connector, a USB connector, or RCA connectors. The input devices may be wireless and connect to the gain control modules through any one of a variety of protocols, including WiMAX, LTE, Bluetooth, Bluetooth Broadcast, GSM, 3G, 4G, 5G, 6G, Zigbee, 60 GHz Wi-Fi, Wi-Fi (e.g., compatible with IEEE 802.11a/b/g/n/ac/ad/af/ah/ai/aj/aq/ax/ay/ba/be), NFC protocols, proprietary wireless connection protocols, and/or any other protocol. Where the connection is wireless, the input devices (and/or their respective transmitters, receivers, or transceivers) and the gain control modules may include a wireless communications interface. The gain control modules may be compatible with any number and type of output devices, such as output device **606**. Output device **606** may receive an output signal, such as signal y(n), from gain control module modules **200***a*-**200***n* for playback, transmittal to another external device, or for further processing. Output device **606** may include, for example, loudspeakers or other audio output devices, an audio mixer, a wireless transmitter, receiver, transceiver, etc.

[0059] As illustrated above with respect to FIG. **6**, gain control module **200** may be compatible with any number and type of input devices, such as microphones. The gain control modules may be configured to operate in concert with auto-mixer **605**. The gain control modules may be equivalent in structure and operation to gain control module **200**. When the gain control modules are utilized with multiple microphones simultaneously and used in combination with auto-mixer **605**, it may be advantageous to estimate the overall effective sensitivity of input devices **100**, **600**, **602**, **604**, so that an equal acoustic sound intensity may be detected equally by each channel of auto-mixer **605**. The gain control modules may be communicatively connected to sensitivity offset controller **608** (hereinafter referred to as "offset controller **608**"). Offset controller **608** may be configured to receive respective audio signals x(n) from devices **100**, **600**, **602**, **604**; respective outputs from respective envelope followers **326** of the gain control modules; and respective indications **336** from respective comparators **332** of the gain control modules.

[0060] Offset controller **608** may be configured to equalize the effective acoustic sensitivity of each respective signal x(n) of each input device **100**, **600**, **602**, **604**. Offset controller **608** may be configured to output respective modified audio signals s(n), equalized for sensitivity, as a secondary input into auto-mixer **605**.

[0061] For example, if there is a significant difference among the estimated noise floors of the signal levels of input devices **100**, **600**, **602**, **604**, it may be advantageous to apply an offset gain to one or more of the signal levels to equalize the effective acoustic sensitivity of each input device **100**, **600**, **602**, **604**. Assuming that acoustic noise (as opposed to electrical noise) is the dominant component of the noise floor of each input, the respective envelope followers of the gain control modules and offset controller **608** may be used to estimate the effective sensitivity of each input device **100**, **600**, **602**, **604**. That is, for example, where input devices **100**, **600**, **602**, **604** are microphones, the acoustic noise component of the noise floor of each input may refer to the combination of the physical sensitivity of the respective microphone transducers (from acoustic pressure to electrical signal) and any gain that may be applied (either in the analog or digital domain) before being input into gain control subsystem **304**.

[0062] The gain control modules may generate indications **336** for each respective input signal of input devices **100**, **600**, **602**, **604**, according to aspects described herein. If the respective indications **336** are "false" for all input devices, offset controller **608** and the respective envelope followers **326** for each of the gain control modules may estimate the acoustic noise floor of each input device **100**, **600**, **602**, **604**. Offset controller **608** may determine the maximum (or minimum) measurement generated by each respective envelope follower **326**. Offset controller **608** may determine the difference in noise floor amplitude between each input and the maximum (or minimum) measurement of the entire set of envelope followers **326**. Offset controller **608** may apply the difference as an offset gain to each respective signal x(n) for each input device **100**, **600**, **602**, **604**. If any of the respective indications **336** are "true," sensitivity controller **608** might not apply a new offset gain and may apply the existing offset gain. If starting from an initial condition, for example, offset controller **608** may apply a gain of 0 dB. Offset controller **608** may be configured to output modified signals s(n), equalized for sensitivity, as a secondary input to auto-mixer **605**.

[0063] An apparatus may comprise one or more features. The apparatus may comprise an audio detector and/or a gain control system. The audio detector may be configured to receive a first audio signal. The audio detector may comprise a first envelope follower configured to generate, from a first audio signal and based on a first time constant, a first amplitude measurement. The audio detector may comprise a second envelope follower configured to generate, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement. The audio detector may comprise a first comparator configured to output a first difference between the first amplitude measurement and the second amplitude measurement. The audio detector may comprise a second comparator configured to generate a first indication of whether the first difference satisfies a first threshold value. The gain control subsystem may comprise a target gain calculation module configured to calculate, based on the first indication, a target gain for the first audio signal. The gain control subsystem may comprise a gain state update module configured to modify, based on the target gain, a gain state associated with the first audio signal. The audio detector may further comprise a third envelope follower configured to generate, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement. The audio detector may further comprise a third comparator configured to output a second difference between the first amplitude measurement and the third amplitude measurement. The audio detector may further comprise a fourth comparator configured to generate a second indication of whether the second difference satisfies a second threshold value. The gain state update module may be further configured to modify, based on the second indication, the gain state associated with the first audio signal. The gain control subsystem may further comprise a fifth comparator configured to output, based at least on the first indication and the second indication, a third indication comprising at least one of: 1) whether to calculate a target gain for the first audio signal; or 2) whether to modify, based on the target gain, the gain state associated with the first audio signal. The gain state update module may be configured to modify the gain state associated with the first audio signal based on a hysteresis of the gain state associated with the first audio signal. The gain state associated with the first audio signal may be modified according to a ramp speed, the ramp speed based on a difference between the target gain and the gain state associated with the first audio signal. At least one of the first envelope follower, the second envelope follower, or the third envelope follower may be configured to measure a first noise floor of the first audio signal. The audio detector may further comprise a fourth envelope follower configured to generate a fourth amplitude measurement of a second noise floor of a second audio signal. The apparatus may further comprise a sensitivity offset controller. The third comparator may be further configured to measure a third difference between the fourth amplitude measurement and the third amplitude measurement. The sensitivity offset controller may be configured to: generate an offset gain based on the third difference; and apply the offset gain to at least one of the first audio signal or the second audio signal. The apparatus may further comprise at least one of a wireless transmitter, a wireless receiver, a wireless transceiver, or a microphone.

[0064] A non-transitory machine-readable storage medium may comprise instructions which, when executed by one or more processors, may cause the one or more processors to perform one or more operations. The one or more operations may comprise generating, from a first audio signal and based on a first time constant, a first amplitude measurement. The one or more operations may comprise generating, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement. The one or more operations may comprise outputting a first difference between the first amplitude measurement and the second amplitude measurement. The one or more operations may comprise generating a first indication of whether the first difference satisfies a first threshold value. The one or more operations may comprise calculating, based on the first indication, a target gain for the first audio signal. The one or more operations may comprise modifying, based on the target gain, a gain state associated with the first audio signal. The one or more operations may comprise generating, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement. The one or more operations may comprise outputting a second difference between the first amplitude measurement and the third amplitude measurement. The one or more operations may comprise generating a second indication of whether the second difference satisfies a second threshold value. The one or more operations may comprise modifying, based on the second indication, the gain state associated with the first audio signal. The one or more operations may comprise outputting, based at least on the first indication and the second indication, a third indication comprising at least one of: 1) whether to calculate a target gain for the first audio signal; or 2) whether to modify, based on the target gain, the gain state associated with the first audio signal. The one or more operations may comprise modifying the gain state associated with the first audio signal based on a hysteresis of the gain state. The one or more operations may comprise modifying the gain state associated with the first audio signal according to a ramp speed, wherein the ramp speed is based on a difference between the target gain and the gain state. The one or more operations may comprise generating, from a second audio signal, a fourth amplitude measurement. The one or more operations may comprise outputting a third difference between the fourth amplitude measurement and the third amplitude measurement. The one or more operations may comprise generating an offset gain based on the third difference. The one or more operations may comprise applying the offset gain to at least one of the first audio signal or the second audio signal.

[0065] A method may comprise one or more operations. The method may comprise generating, by a computing device, a first amplitude measurement of a first audio signal and based on a first time constant. The method may comprise generating, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement. The method may comprise outputting a first difference between the first amplitude measurement and the second amplitude measurement. The method may comprise generating a first indication of whether the first difference satisfies a first threshold value. The method may comprise calculating, based on the first indication, a target gain for the first audio signal. The method may comprise modifying, based on the target gain, a gain state associated with the first audio signal. The method may comprise generating, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement. The method may comprise outputting a second difference between the first amplitude measurement and the third amplitude measurement. The method may comprise generating a second indication of whether the second difference satisfies a second threshold value. The method may comprise modifying, based on the second indication, the gain state associated with the first audio signal. The method may further comprise: generating, from a second audio signal, a fourth amplitude measurement; outputting a third difference between the fourth amplitude measurement and the third amplitude measurement; generating an offset gain based on the third difference; and/or applying the offset gain to at least one of the first audio signal or the second audio signal. The modifying may further comprise modifying the gain state associated with the first audio signal based on a hysteresis of the gain state associated with the first audio signal. The modifying may further comprise modifying the gain state associated with the first audio signal according to a ramp speed, wherein the ramp speed is based on a difference between the target gain and the gain state associated with the first audio signal. The method may be performed by an apparatus described herein. A system may comprise the apparatus and one or more other elements described herein.

[0066] In the foregoing specification, the present disclosure has been described with reference to specific exemplary examples thereof. Although the invention has been described in terms of a preferred example, those skilled in the art will recognize that various modifications, examples or variations of the invention can be practiced within the spirit and scope of the invention as set forth in the appended claims. The specification and drawings are, therefore, to be regarded in an illustrative rather than restrictive sense. Accordingly, it is not intended that the invention be limited except as may be necessary in view of the appended claims.

Claims

1. An apparatus comprising: an audio detector configured to receive a first audio signal, wherein the audio detector comprises: a first envelope follower configured to generate, from a first audio signal and based on a first time constant, a first amplitude measurement; a second envelope follower configured to generate, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement; a first comparator configured to output a first difference between the first amplitude measurement and the second amplitude measurement; and a second comparator configured to generate a first indication of whether the first difference satisfies a first threshold value; and a gain control subsystem comprising: a target gain calculation module configured to calculate, based on the first indication, a target gain for the first audio signal; and a gain state update module configured to modify, based on the target gain, a gain state associated with the first audio signal.
2. The apparatus of claim 1, wherein: the audio detector further comprises: a third envelope follower configured to generate, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement; a third comparator configured to output a second difference between the first amplitude measurement and the third amplitude measurement; and a fourth comparator configured to generate a second indication of whether the second difference satisfies a second threshold value; and wherein the gain state update module is further configured to modify, based on the second indication, the gain state associated with the first audio signal.
3. The apparatus of claim 2, wherein the gain control subsystem further comprises a fifth comparator configured to output, based at least on the first indication and the second indication, a third indication comprising at least one of: whether to calculate a target gain for the first audio signal; or whether to modify, based on the target gain, the gain state associated with the first audio signal.
4. The apparatus of claim 2, wherein at least one of the first envelope follower, the second envelope follower, or the third envelope follower is configured to measure a first noise floor of the first audio signal.
5. The apparatus of claim 4, wherein the audio detector further comprises a fourth envelope follower configured to generate a fourth amplitude measurement of a second noise floor of a second audio signal.
6. The apparatus of claim 5, further comprising a sensitivity offset controller, wherein: the third comparator is further configured to: measure a third difference between the fourth amplitude measurement and the third amplitude measurement; and the sensitivity offset controller is configured to: generate an offset gain based on the third difference; and apply the offset gain to at least one of the first audio signal or the second audio signal.
7. The apparatus of claim 1, wherein the gain state update module is configured to modify the gain state associated with the first audio signal based on a hysteresis of the gain state associated with the first audio signal.
8. The apparatus of claim 7, wherein the gain state associated with the first audio signal is modified according to a ramp speed, and wherein the ramp speed is based on a difference between the target

gain and the gain state associated with the first audio signal.
9. The apparatus of claim 1, further comprising at least one of a wireless transmitter, a wireless receiver, a wireless transceiver, or a microphone.
10. A non-transitory machine-readable storage medium comprising instructions that, when executed by one or more processors, cause the one or more processors to: generate, from a first audio signal and based on a first time constant, a first amplitude measurement; generate, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement; output a first difference between the first amplitude measurement and the second amplitude measurement; generate a first indication of whether the first difference satisfies a first threshold value; calculate, based on the first indication, a target gain for the first audio signal; and modify, based on the target gain, a gain state associated with the first audio signal.
11. The non-transitory machine-readable storage medium of claim 10, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: generate, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement; output a second difference between the first amplitude measurement and the third amplitude measurement; generate a second indication of whether the second difference satisfies a second threshold value; and modify, based on the second indication, the gain state associated with the first audio signal.
12. The non-transitory machine-readable storage medium of claim 11, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to output, based at least on the first indication and the second indication, a third indication comprising at least one of: whether to calculate a target gain for the first audio signal; or whether to modify, based on the target gain, the gain state associated with the first audio signal.
13. The non-transitory machine-readable storage medium of claim 11, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to: generate, from a second audio signal, a fourth amplitude measurement; output a third difference between the fourth amplitude measurement and the third amplitude measurement; generate an offset gain based on the third difference; and apply the offset gain to at least one of the first audio signal or the second audio signal.
14. The non-transitory machine-readable storage medium of claim 10, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to modify the gain state associated with the first audio signal based on a hysteresis of the gain state.
15. The non-transitory machine-readable storage medium of claim 14, wherein the instructions, when executed by the one or more processors, further cause the one or more processors to modify the gain state associated with the first audio signal according to a ramp speed, wherein the ramp speed is based on a difference between the target gain and the gain state.
16. A method comprising: generating, by a computing device, a first amplitude measurement of a first audio signal and based on a first time constant; generating, from the first audio signal and based on a second time constant different from the first time constant, a second amplitude measurement; outputting a first difference between the first amplitude measurement and the second amplitude measurement; generating a first indication of whether the first difference satisfies a first threshold value; calculating, based on the first indication, a target gain for the first audio signal; and modifying, based on the target gain, a gain state associated with the first audio signal.
17. The method of claim 16, further comprising: generating, from the first audio signal and based on a third time constant different from the first time constant and second time constant, a third amplitude measurement; outputting a second difference between the first amplitude measurement and the third amplitude measurement; and generating a second indication of whether the second difference satisfies a second threshold value; and modifying, based on the second indication, the gain state associated with the first audio signal.
18. The method of claim 17, further comprising: generating, from a second audio signal, a fourth

amplitude measurement; outputting a third difference between the fourth amplitude measurement and the third amplitude measurement; generating an offset gain based on the third difference; and applying the offset gain to at least one of the first audio signal or the second audio signal.

19. The method of claim 16, wherein the modifying further comprises modifying the gain state associated with the first audio signal based on a hysteresis of the gain state associated with the first audio signal.

20. The method of claim 19, wherein the modifying further comprises modifying the gain state associated with the first audio signal according to a ramp speed, wherein the ramp speed is based on a difference between the target gain and the gain state associated with the first audio signal.